

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester VI Examination, 2022-23

Paper No. STA/STB-601: Numerical Analysis

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks. Symbols have their usual meanings.*

Question 1.

- (a) Find a polynomial that fits the data in the following table:

x	0	1	2	3
$f(x)$	2	6	12	20

Also prove that $\Delta^n(x^n) = n!$ where n is a positive integer. [7]

- (b) Show that the difference operator Δ is a linear operator. [7]

Question 2.

- (a) State and prove Newton's backward interpolation formula and give its importance. [7]

- (b) State and prove Bessel's interpolation formula. [7]

Question 3.

- (a) Show that divided differences are symmetric in their arguments. [7]

- (b) Derive the error term in the Lagrange interpolation formula. [7]

Question 4.

- (a) Define a homogeneous linear difference equation. Discuss its solution under the different cases of roots of the characteristic equation. [7]

- (b) Let $\phi(E) = a_0E^n + a_1E^{n-1} + \dots + a_n$. Prove that:

(i) $\phi(E) \beta^k = \phi(\beta) \beta^k$

- (ii) The particular solution of the equation $\phi(E) \beta^k = p^k$ is given by $\frac{p^k}{\phi(p)}$, provided $\phi(p) \neq 0$.

Also solve the difference equation $y_{k+2} - 3y_{k+1} + 2y_k = 4^k$. [7]

Question 5.

- (a) Prove the fundamental theorem of sum calculus (summation by parts). [7]
- (b) Find $\Delta^{-1}(x \cdot a^x)$, $a \neq 1$. [7]
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Question 6.

- (a) Prove that $1^3 + 2^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$. [7]
- (b) Discuss the problem of numerical differentiation. Suppose that $x_0, x_1, \dots, x_n$ are distinct numbers in some interval J and the function f is $(n+1)$ times differentiable. Obtain an $(n+1)$ -point formula to approximate $f'(x_j)$ using Lagrange's formula. [7]
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Question 7.

- (a) Determine the value of h that will ensure an approximation error of less than 0.00002 when approximating $\int_0^\pi \sin(x) dx$ using:
- (i) Composite Trapezoidal rule
 - (ii) Composite Simpson's 1/3 rule
- [7]
- (b) Show that Simpson's 1/3 rule for numerical integration is more accurate than Simpson's 3/8 rule. [7]
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Question 8.

- (a) Discuss the Runge-Kutta formula for finding a numerical solution of differential equations. Write the expression for the fourth-order Runge-Kutta formula. [7]
- (b) Discuss the improved Euler's method. Taking $h = 0.05$, approximate the solution of the initial value problem $\frac{dy}{dx} = x + y$, $y(0) = 1$, at $x = 0.05$ (execute only the first iteration). [7]
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